Zain Naqvi

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Education

McMaster University — Hamilton, ON
Bachelors in Computer Engineering CO-OP - GPA: 3.82 — *Sept. 2024 – May 2029*

Experience

Lead - Electrical Engineering Sub-Team — September 2025 - Present
McMaster RoboSub — *Hamilton, Ontario*

- Built a hardware-in-the-loop DVL emulator on a Teensy 4.1 using a custom C++ (OOP) library that replicates the Nortek Nucleus 1000's Ethernet interface for testing
- Designing the AUV's acoustics board in Altium and building DIY hydrophones from piezoelectric sensors to triangulate underwater acoustic signals for localization
- As Team Lead, rebuilt the team and secured $6,000 in Faculty funding. Made a 12S - 5V at 3A Buck Converter on Altium

Projects

STM32U5 Secure Bootloader | *C, ARM Cortex-M33 (ARMv8-M), STM32U585, Bare-Metal, Secure Boot* — *In progress*

- Writing a secure bootloader for the STM32U585 (Cortex-M33) from the RM0456 manual with no HAL. Custom linker script and a bootloader-to-application handoff that relocates VTOR, sets the MSP, then branches, with nothing allowed between the stack switch and the branch. Verified across a cold reset
- Wrote the embedded flash driver at register level: two-key unlock, page erase, and quad-word programming against the part's 128-bit plus 9-bit ECC granularity. Provoked PGAERR and PROGERR deliberately to confirm the error paths catch them, and traced a stale post-erase read-back to the ICACHE, now invalidated before every read after a write
- Split the A/B slots across both flash banks so an erase to the staging bank never stalls fetch on the running bank, with log-structured metadata recoverable after power loss mid-write. Image hashed by the HASH peripheral over GPDMA

Linux PCI Device Driver in QEMU | *C, QEMU, Linux Kernel, PCI, MMIO, MSI, DMA*

- Benchmarked programmed I/O against DMA across 11 transfer sizes with kernel-side ktime timestamps and got a flat 100 ms floor on every DMA size. Traced it to a hardcoded timer in QEMU's device model rather than reporting it as real transfer cost
- Built the driver for QEMU's EDU device against the emulated hardware: ID-table match, probe/remove, BAR 0 mapped with ioremap, and a misc character device at /dev/edu. Moved the read path off busy-polling onto MSI interrupt-driven I/O, then added 4 KB coherent DMA behind an ioctl with a shared ABI header and a 28-bit DMA mask

Bare-Metal RTOS Kernel | *C, ARM Thumb Assembly, Cortex-M4*

- Built a preemptive RTOS from scratch on an ARM Cortex-M4 MSP432E401Y MCU with no HAL or vendor libraries. Context switcher written in ARM Thumb assembly using SysTick and PendSV exceptions, with isolated Process Stack Pointers and manually forged 16-word initial stack frames per task
- Implemented binary semaphores with priority-ordered wait queues, mutexes with priority inheritance, and a first-fit heap with block splitting and stack overflow detection. Debugged a real priority inversion and a PendSV ordering race on live hardware over CMSIS-DAP

Multi-Node CAN Bus Network & Intrusion Detection System | *C, ARM Cortex-M4, CAN Bus, Embedded Security*

- Drove both on-chip CAN controllers on an MSP432E401Y as independent ECUs across a twisted-pair bus with external SN65HVD230 transceivers, configuring every register by hand including bit timing for 500 kbps. Implemented a firmware IDS with five real-time detectors covering over-range, too-fast frames, missing frames, impossible jumps, and rogue message IDs
- Measured it with the Cortex-M4 DWT cycle counter: under 8 microseconds single-frame detection latency, 2.8 microsecond worst-case interrupt handler, near-zero false positives across boundary traffic, and 98% CPU headroom. Each detector validated on live hardware through a UART fault-injection framework

Technical Skills

Languages: C, C++, ARM Thumb Assembly, Verilog, Python
Firmware and Boot: Bare-metal C, RTOS, secure boot and chain of trust, embedded flash programming, custom linker scripts, vector table relocation, DMA, ARM Cortex-M4 and Cortex-M33 (ARMv8-M), NVIC / SysTick / PendSV
Debug and Bring-Up: SWD, CMSIS-DAP, STLINK-V3E, GDB, Keil MDK, STM32CubeIDE, DWT cycle profiling, logic analyzer (Digilent AD3), root-causing firmware and hardware issues from bench captures
Linux and Emulation: Linux kernel module development, PCI subsystem, MMIO, MSI interrupt handling, DMA API, ioctl interfaces, character devices, QEMU
Hardware and Tools: Schematic and PCB design (Altium), Verilog RTL and testbench simulation, PSpice, sensor integration, I2C / SPI / UART / CAN, Git, Linux